Group Homework 3: The Impossibility of Simultaneous Calibration and Error-Rate Parity

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Declaration on the Use of Generative AI

In this course, we use generative artificial intelligence (GenAI) tools such as Claude and ChatGPT to better understand the concept of calibration, error-rate parity, and the Impossibility Theorem in the context of Responsible ML and Algorithmic Bias Measurement. We declare that this was done with integrity and transparency, and that some aspects of GenAI were used as learning aids during brainstorming, preparatory research, and exploratory analysis. Therefore, the final report is a product of our own understanding and synthesis of the concepts, and we have not directly copied any content generated by GenAI. We have cited any sources or references that informed our work, and we have ensured that the report reflects our own critical thinking and insights on the topic.

Problem Given

Let h be a binary classifier, $S \in [0, 1]$ a score, $Y \in \{0, 1\}$ the outcome, and $A \in \{0, 1\}$ a protected attribute.

(a) State the conditions for calibration, FPR parity, and FNR parity.

(b) Prove that when $P(Y = 1 \mid A = 0) \neq P(Y = 1 \mid A = 1)$, no non-trivial classifier satisfies all three simultaneously.

(c) Apply the impossibility result to the COMPAS empirical findings from Lecture 01. Which criterion did COMPAS satisfy? Which did it violate?

(d) Propose a governance rule: under what circumstances should an organization prioritize calibration over FPR parity? Justify using the burden-shifting framework.

The Solutions

The solution to the problem is structured into four parts corresponding to the sub-questions (a), (b), (c), and (d).

(a) Conditions for Calibration, FPR Parity, and FNR Parity

Calibration (Predictive Parity)

Calibration, or predictive parity, states that for individuals who receive the same score s, the proportion who truly reoffend ($Y = 1$) must be the same across groups, regardless of protected attribute. For every score value $s \in [0, 1]$:

$$P(Y = 1 \mid S = s, A = 0) = P(Y = 1 \mid S = s, A = 1)$$

The score is equally meaningful across all groups. A score of 0.8 should indicate an 80% probability of $Y = 1$ for both $A = 0$ and $A = 1$. This is the criterion Northpointe invoked to defend COMPAS.

FPR Parity

Parity in the False Positive Rate states that false positives – individuals who do not reoffend ($Y = 0$) but are labeled high-risk ($h = 1$) – should occur at the same rate across groups. Given our binary classifier h:

$$P(h = 1 \mid Y = 0, A = 0) = P(h = 1 \mid Y = 0, A = 1)$$

For people who truly have a zero outcome (do not reoffend), the classifier should produce the same false alarm rate regardless of protected attribute.

FNR Parity

Parity in the False Negative Rate states that false negatives – individuals who do reoffend ($Y = 1$) but are labeled low-risk ($h = 0$) – should occur at the same rate across groups:

$$P(h = 0 \mid Y = 1, A = 0) = P(h = 0 \mid Y = 1, A = 1)$$

For people who truly have a one outcome (do reoffend), the classifier should miss them – labeling them low-risk – at the same rate regardless of protected attribute. Note that FNR parity is equivalent to TPR parity since $FNR_a = 1 - TPR_a$.

(b) Proof: Impossibility When Base Rates Differ

Claim: When $P(Y = 1 \mid A = 0) \neq P(Y = 1 \mid A = 1)$, no non-trivial classifier can simultaneously satisfy calibration, FPR parity, and FNR parity.

Notation

For each group $a \in \{0, 1\}$, define the base rate $\pi_a := P(Y = 1 \mid A = a)$ and the following per-group quantities:

- true positive rate: $TPR_a := P(h = 1 \mid Y = 1, A = a)$ – the probability that an individual is labeled high-risk, given they reoffend and have the protected attribute a
- false positive rate: $FPR_a := P(h = 1 \mid Y = 0, A = a)$ – the probability that an individual is labeled high-risk, given they do not reoffend and have the protected attribute a
- false negative rate: $FNR_a := P(h = 0 \mid Y = 1, A = a) = 1 - TPR_a$ – the probability that an individual is labeled low-risk, given they do reoffend
- positive predictive value: $PPV_a := P(Y = 1 \mid h = 1, A = a)$ – the probability that an individual reoffends, given they were labeled high-risk and have the protected attribute a

Symbol	Definition	Interpretation
π_a	$P(Y = 1 \mid A = a)$	Base rate: probability of reoffending in group a

Symbol	Definition	Interpretation
TPR_a	$P(h=1 \mid Y=1, A=a)$	Probability labeled high-risk given true reoffending
FPR_a	$P(h=1 \mid Y=0, A=a)$	Probability labeled high-risk given true non-reoffending
FNR_a	$P(h=0 \mid Y=1, A=a) = 1 - TPR_a$	Probability labeled low-risk given true reoffending
PPV_a	$P(Y=1 \mid h=1, A=a)$	Probability of true reoffending given labeled high-risk

Fairness Conditions

Under the three fairness criteria:

- FPR parity (f): $FPR_0 = FPR_1$ – the false positive rate should be equal among groups
- FNR parity (t): $FNR_0 = FNR_1 \Rightarrow TPR_0 = TPR_1$ – the false negative (and true positive) rate should be equal among groups
- Calibration (predictive parity): $PPV_0 = PPV_1$ – the calibration should be equal among groups

Assume all three conditions hold jointly. We define:

$$t := TPR_0 = TPR_1$$

$$f := FPR_0 = FPR_1$$

For the classifier to be non-trivial and non-perfect, we require:

$$0 < t < 1$$

$$0 < f < 1$$

Express the Selection Rate per Group

The probability of predicting positive in group a follows by the law of total probability:

$$P(h=1 \mid A=a) = P(h=1 \mid Y=1, A=a)P(Y=1 \mid A=a) + P(h=1 \mid Y=0, A=a)P(Y=0 \mid A=a)$$

$$= TPR_a \pi_a + FPR_a(1-\pi_a) = t\pi_a + f(1-\pi_a)$$

The probability that an individual was labeled high-risk, given that they have the protected attribute a, is equal to the true positive rate of group a multiplied by the base rate of group a, plus the false positive rate of group a multiplied by one minus the base rate of group a.

Derive PPV Using Base Rate and Error Rates

By Bayes' theorem, the positive predictive value for group a is:

$$PPV_a = P(Y=1 \mid h=1, A=a) = \frac{P(h=1 \mid Y=1, A=a)P(Y=1 \mid A=a)}{P(h=1 \mid A=a)} = \frac{TPR_a \pi_a}{t\pi_a + f(1-\pi_a)}$$

The positive predictive value for both groups equals the true positive rate multiplied by the base rate, divided by the true positive rate multiplied by the base rate plus the false positive rate multiplied by one minus the base rate.

Apply Calibration (Predictive Parity) to Relate Base Rates

Using $t = TPR_a$ and $f = FPR_a$ for both groups, calibration requires $PPV_0 = PPV_1$:

$$\frac{t\pi_0}{t\pi_0 + f(1-\pi_0)} = \frac{t\pi_1}{t\pi_1 + f(1-\pi_1)}$$

Assuming a non-trivial, non-perfect classifier ($0 < t < 1$, $0 < f < 1$), we cross-multiply to obtain:

$$t\pi_0[t\pi_1 + f(1-\pi_1)] = t\pi_1[t\pi_0 + f(1-\pi_0)]$$

Expanding both sides:

$$t^2\pi_0\pi_1 + tf\pi_0(1-\pi_1) = t^2\pi_1\pi_0 + tf\pi_1(1-\pi_0)$$

The $t^2\pi_0\pi_1$ terms cancel from both sides:

$$tf\pi_0(1-\pi_1) = tf\pi_1(1-\pi_0)$$

Since $t > 0$ and $f > 0$, we divide both sides by $tf > 0$:

$$\pi_0(1-\pi_1) = \pi_1(1-\pi_0)$$

Expanding both sides:

$$\pi_0 - \pi_0\pi_1 = \pi_1 - \pi_0\pi_1$$

The $\pi_0\pi_1$ terms cancel from both sides, leaving:

$$\pi_0 = \pi_1$$

Contradiction and Conclusion

Thus, if calibration, FPR parity, and FNR parity all hold for a non-trivial, non-perfect classifier, the base rates must be equal:

$$P(Y = 1 \mid A = 0) = P(Y = 1 \mid A = 1)$$

Any non-trivial classifier that satisfies all three (calibration, FPR parity, and FNR parity) must have $\pi_0 = \pi_1$, which is a contradiction. Therefore, when $P(Y = 1 \mid A = 0) \neq P(Y = 1 \mid A = 1)$, no non-trivial classifier can satisfy calibration, FPR parity, and FNR parity simultaneously.

(c) Application to COMPAS – Lecture 01 Empirical Findings

Background

The COMPAS risk-assessment tool was deployed in Broward County, FL, and scored approximately 7,000 defendants for predicted recidivism. The ProPublica investigation (2016) found the following empirical pattern:

- $FPR_{Black} > FPR_{White}$
- $FNR_{White} > FNR_{Black}$

Which Criterion Did COMPAS Satisfy?

Calibration – approximately satisfied. Northpointe, COMPAS's developer, demonstrated that the scores were approximately equally predictive at each score level across racial groups:

$$P(Y = 1 \mid S = s, A = Black) \approx P(Y = 1 \mid S = s, A = White)$$

A score of 7, for example, corresponded to roughly the same empirical recidivism probability regardless of race. This is the criterion Northpointe invoked as its primary defense, and the Lecture 01 live-coding results confirmed approximate calibration across groups.

Which Criteria Did COMPAS Violate?

FPR parity – violated. Black defendants who did not reoffend were labeled high-risk at a significantly higher rate than White defendants who did not reoffend. From the ProPublica analysis and Lecture 01 replication:

- $FPR_{Black} \approx 0.45$
- $FPR_{White} \approx 0.24$

FNR parity – violated. White defendants who did reoffend were more frequently labeled low-risk (missed by the model) than Black defendants who did reoffend:

- $FNR_{White} > FNR_{Black}$

Why the Theorem Predicts This Outcome

These findings are not a flaw in COMPAS's design in isolation – they are a direct consequence of the Impossibility Theorem proved in Part (b). Base recidivism rates differ between the two groups in Broward County:

$$\pi_{Black} = P(Y = 1 \mid A = Black) > P(Y = 1 \mid A = White) = \pi_{White}$$

Because $\pi_{Black} \neq \pi_{White}$, Part (b) proves it is mathematically impossible to simultaneously achieve calibration and FPR/FNR parity. COMPAS's designers implicitly chose calibration – and the resulting FPR and FNR disparities are the unavoidable arithmetic consequence of that choice applied to groups with different base rates.

The ProPublica–Northpointe dispute was not about whether the model was broken: it was a dispute about which fairness criterion to prioritize, and both sides were correct within their chosen criterion. This is precisely the governance question Part (d) addresses.

(d) Governance Rule – When to Prioritize Calibration over FPR Parity

The Burden-Shifting Framework

Under *Griggs v. Duke Power Co.* (1971) and subsequent EEOC guidance, an organization facing a disparate impact claim navigates three steps:

Step	Actor	Obligation
1	Plaintiff	Demonstrate a prima facie statistical disparity (e.g., AIR < 0.80 or $p < 0.05$ on a two-proportion z-test for FPR)
2	Defendant	Demonstrate business necessity – the practice is job-related and consistent with legitimate operational need
3	Plaintiff	Demonstrate that a less discriminatory alternative exists that meets the same business necessity

Proposed Governance Rule

An organization may defensibly prioritize calibration over FPR parity when all three of the following conditions are jointly met:

- **Condition 1 – Accurate individual risk prediction is the primary operational purpose.** The model is used to allocate resources, set supervision levels, or inform (not dictate) decisions based on genuine likelihood of outcome – not as a binary gate to liberty or opportunity. Where false positives carry severe irreversible consequences (e.g., pretrial detention, employment rejection), FPR parity is the more appropriate criterion because the harm of over-classification is asymmetric and concentrated on the disadvantaged group.
- **Condition 2 – Differential base rates are empirically documented and attributable to factors independent of protected status.** The organization must demonstrate, with audited data and independent review, that $\pi_0 \neq \pi_1$ reflects genuine outcome differences in the deployment population – not historical discrimination encoded in training labels. If differential base rates are themselves products of past discriminatory practices (e.g., racially biased arrest patterns), choosing calibration launders that discrimination into a new algorithmic form and likely fails the Step 2 business necessity test.
- **Condition 3 – No less discriminatory alternative achieves equivalent predictive accuracy.** The organization must actively search for and document alternative models, thresholds, or feature sets that reduce FPR disparity without material loss of calibration

or predictive utility. Calibration may only be invoked as the governing criterion if the organization can show at Step 2 that no such alternative exists, preempting the plaintiff's Step 3 claim. This search must be documented, dated, and repeated periodically as new data becomes available.

When FPR Parity Should Be Prioritized Instead

Calibration should not dominate when:

- The decision gates access to a fundamental liberty interest (bail, parole, sentencing) – the asymmetric harm of a false positive on a non-recidivist justifies FPR parity as the binding constraint, even at the cost of some miscalibration.
- The affected group already experiences systemic disadvantage – layering a calibrated but FPR-unequal algorithm on top of existing structural inequity amplifies rather than ameliorates harm, and likely triggers disparate impact liability regardless of business necessity.
- A statistically significant FPR disparity ($p < 0.05$) is accompanied by a practically significant effect (AIR < 0.80) – both thresholds triggered simultaneously is the clearest case for remediation under EEOC guidance.

Documentation Requirements

Regardless of which criterion is prioritized, organizations must:

- Record the fairness criterion selected, the rationale, and the date of the decision.
- Retain the statistical tests and effect sizes that triggered the disparate impact review.
- Commission independent audits at regular intervals or following model updates.
- Build remediation workflows for cases where new data shifts the base-rate differential or introduces a less-discriminatory alternative.

Failure to document this choice does not make the impossibility go away – it simply transfers legal liability from a defensible policy decision to an indefensible omission.

Conclusion

We have shown that calibration, FPR parity, and FNR parity are three distinct and in general mutually incompatible fairness criteria. When base rates differ across groups – as they do in the Broward County COMPAS dataset – the Impossibility Theorem (Chouldechova, 2017) proves that no non-trivial classifier can satisfy all three simultaneously.

Applying this result to COMPAS: the system was approximately calibrated (Northpointe's defense) but violated both FPR and FNR parity (ProPublica's finding). Both sides of the dispute were mathematically correct within their chosen criterion. The disagreement was therefore not empirical but normative – a value choice about which errors matter most, for whom, and under what legal standard.

The governance rule proposed in Part (d) operationalizes this choice through the burden-shifting framework: calibration is defensible as the primary criterion only when accurate risk prediction is the legitimate business purpose, differential base rates are independently documented, and no less-discriminatory alternative exists. In high-stakes liberty contexts such as criminal justice, the asymmetric harm of false positives on non-recidivists – concentrated on already-disadvantaged groups – typically means FPR parity should take precedence.

Ultimately, the Impossibility Theorem is not a reason for inaction. It is a mandate for transparency: document the criterion chosen, quantify the trade-off accepted, monitor it over time, and be prepared to justify it in court.

References

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DNSC 6330 Lecture 03: Algorithmic Bias Measurement (course material on AIR, FPR/FNR parity, COMPAS, and the burden-shifting framework).